

Defect Detection Results Report

Specimen: octet_9x9x9_defects

Source JSON: octet_9x9x9_defects.json

Generated: 2026-07-28T16:04:53-07:00

Overall condition rating: Good

Registered Design Summary

Feature	Value	Source
Specimen ID	210127_Brian_Tran_strut_lattices_0point5dash1 1 Slices	metadata/path
Registered JSON filename	210127_Brian_Tran_strut_lattices_0point5dash1 1 Slices.json	discovered/explicit
Registered JSON path	C:\Users\kendr\Desktop\DSC 2026\Group\Inl_data_science_challenge_2026-main\data\missing_struts\registered_jsons\210127_Brian_Tran_strut_lattices_0point5dash1 1 Slices.json	discovered/explicit
Lattice type	octet unit-cell graph	metadata/schema
Total struts	18468	struts
Total junctions	10206	junctions
Total octet lattice cells	729	unit_cells
Unit-cell dimensions	9 × 9 × 9	structured indices/metadata
Coordinate bounds	x=58.7606 to 773.745; y=48.5686 to 764.939; z=24.4953 to 737.846	junctions.*.position
Minimum strut thickness	0.1	calculated
Maximum strut thickness	0.1	calculated
Average strut thickness	0.1	calculated
Median strut thickness	0.1	calculated
Unique thickness values	1	calculated

Evaluation Coverage

Metric	Value
Registered struts	18468
Evaluated struts	18468
Duplicate registered IDs	0

Metric	Value
Duplicate evaluation IDs	0

- ID-level coverage is unavailable because one or both strut collections lack reliable identifiers.

Executive Summary

The supplied result contains 18,468 struts. 17,253 (93.42%) are intact and 1,215 (6.58%) are classified as missing, broken/disconnected, bent, or thinned.

This condition rating is based only on detected defect frequency. It is not a structural safety determination.

Metric	Value
Total Struts	18,468
Intact Percentage	93.42%
Total Defective Percentage	6.58%
Severe Defect Percentage	3.05%
Most Common Defect	Missing
Condition Rating	Good

Source Information

Input JSON path: C:\Users\kendr\Desktop\DSC

2026\Group\lInl_data_science_challenge_2026-main\analysis\defect_detection\sample_output\octet_9x9x9_defects.json

Detected data structure: strut-level records at `struts`

Meta.Volume Shape Zyx: [761, 815, 837]

Defect Distribution

Strut Condition	Count	Percentage
Intact	17,253	93.42%
Missing	410	2.22%
Broken (source: disconnected)	153	0.83%
Bent	365	1.98%
Thinned (source: thin)	287	1.55%
Total defective	1,215	6.58%
Severe defects	563	3.05%

Key Findings

- Most struts are intact.
- Missing is the most frequent nonzero defect; Broken is the least frequent nonzero defect.

- Missing and broken/disconnected members contribute 563 severe defects.

Data Validation and Warnings

- Expected category `uncertain` is absent or has zero records.

Limitations

- The report depends on the supplied defect-detection output.
- The reporting agent does not independently verify the 3D CT scan.
- Defect counts do not describe the structural importance of individual struts.
- Defect percentages do not determine structural safety.
- Location-aware engineering analysis and FEM would be required for structural conclusions.

Recommended Next Steps

- Inspect missing and broken/disconnected struts in the original CT volume.
- Use the endpoint coordinates to visualize defects and investigate connected clusters.
- Compare predictions with the registered STL or design JSON where available.
- Perform structural analysis before drawing engineering or safety conclusions.

Spatial Defect Concentration Analysis

Grouping method: coordinate grid with 3x3x3 bins

Rank	Region	Location	Struts	Defects	Defect %	Ratio	Dominant
1	X-low / Y-high / Z-high	Grid region X-low / Y-high / Z-high; midpoint ranges x=64.27-284.04, y=521.42-754.12, z=503.40-742.11 (source-coordinate units)	685	97	14.16%	2.15	Missing
2	X-high / Y-high / Z-high	Grid region X-high / Y-high / Z-high; midpoint ranges x=531.88-765.50, y=521.34-751.75, z=518.13-744.62 (source-coordinate units)	709	84	11.85%	1.80	Missing
3	X-middle / Y-high / Z-high	Grid region X-middle / Y-high / Z-high; midpoint ranges x=297.65-531.07, y=523.19-752.62, z=517.74-745.35 (source-coordinate units)	688	76	11.05%	1.68	Missing

Detailed Regional Findings

X-low / Y-high / Z-high

Location: Grid region X-low / Y-high / Z-high; midpoint ranges x=64.27-284.04, y=521.42-754.12, z=503.40-742.11

(source-coordinate units)

Defects: 97/685 (14.16%); ratio 2.15; severe 8.91%; dominant Missing.

Counts: missing=44, broken=17, bent=13, thinned=23.

Flagged because: concentration ratio 2.15 meets the 1.50 relative threshold

X-high / Y-high / Z-high

Location: Grid region X-high / Y-high / Z-high; midpoint ranges x=531.88-765.50, y=521.34-751.75,

z=518.13-744.62 (source-coordinate units)

Defects: 84/709 (11.85%); ratio 1.80; severe 7.76%; dominant Missing.

Counts: missing=37, broken=18, bent=16, thinned=13.

Flagged because: concentration ratio 1.80 meets the 1.50 relative threshold

X-middle / Y-high / Z-high

Location: Grid region X-middle / Y-high / Z-high; midpoint ranges x=297.65-531.07, y=523.19-752.62,

z=517.74-745.35 (source-coordinate units)

Defects: 76/688 (11.05%); ratio 1.68; severe 7.41%; dominant Missing.

Counts: missing=36, broken=15, bent=20, thinned=5.

Flagged because: concentration ratio 1.68 meets the 1.50 relative threshold

Exterior Faces of the Large Cube

This analysis applies only to the six exterior faces of the complete object. Internal volumetric-region boundaries are not treated as physical faces.

Rank	Face	Struts	Defects	Defect %	Severe %	Dominant
1	Back	1650	382	23.15%	21.33%	Missing
2	Top	1547	149	9.63%	3.17%	Bent
3	Front	1638	130	7.94%	1.89%	Bent
4	Bottom	1480	106	7.16%	2.50%	Bent
5	Right	1633	90	5.51%	0.98%	Thinned
6	Left	1636	89	5.44%	0.92%	Thinned

Nine Volumetric Regions

Volume grid: 3 x 3 x 1. Total volumetric regions: 9. No internal region boundary is analyzed as a physical face.

Rank	Volume Region	Bounds	Struts	Defects	Defect %	Ratio	Dominant
1	VOLUME_X_LOW_Y_HIGH	x=61.80-296.82, y=521.32-756.89, z=17.33-745.45	2087	220	10.54%	1.60	Missing
2	VOLUME_X_HIGH_Y_HIGH	x=531.85-766.88, y=521.32-756.89, z=17.33-745.45	2148	215	10.01%	1.52	Missing
3	VOLUME_X_MIDDLE_Y_HIGH	x=296.82-531.85, y=521.32-756.89, z=17.33-745.45	2091	179	8.56%	1.30	Missing
4	VOLUME_X_MIDDLE_Y_LOW	x=296.82-531.85, y=50.17-285.74, z=17.33-745.45	1998	127	6.36%	0.97	Thinned

Rank	Volume Region	Bounds	Struts	Defects	Defect %	Ratio	Dominant
5	VOLUME_X_HIGH_Y_LOW	x=531.85-766.88, y=50.17-285.74, z=17.33-745.45	2031	122	6.01%	0.91	Bent
6	VOLUME_X_LOW_Y_LOW	x=61.80-296.82, y=50.17-285.74, z=17.33-745.45	2066	114	5.52%	0.84	Bent
7	VOLUME_X_LOW_Y_MIDDLE	x=61.80-296.82, y=285.74-521.32, z=17.33-745.45	1994	82	4.11%	0.63	Bent
8	VOLUME_X_MIDDLE_Y_MIDDLE	x=296.82-531.85, y=285.74-521.32, z=17.33-745.45	2026	81	4.00%	0.61	Bent
9	VOLUME_X_HIGH_Y_MIDDLE	x=531.85-766.88, y=285.74-521.32, z=17.33-745.45	2027	75	3.70%	0.56	Bent

Highly Concentrated Volumetric Regions

VOLUME_X_LOW_Y_HIGH: 220/2087 defects (10.54%), ratio 1.60, dominant Missing. Counts: missing=130, broken=24, bent=23, thinned=43. Flagged by relative concentration threshold.

VOLUME_X_HIGH_Y_HIGH: 215/2148 defects (10.01%), ratio 1.52, dominant Missing. Counts: missing=122, broken=29, bent=30, thinned=34. Flagged by relative concentration threshold.

Comparison of Surface and Volume Findings

Surface and volume results are reported as separate normalized analyses. No internal volumetric-region face was created or evaluated.